# JHNSRE

## The JHU Near-Space Radiation Experiment

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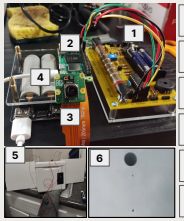
*With thanks to Fiore True Value Hardware, and the students of Bishop Guilfoyle Academy.*

Radiation permeates the atmosphere at all altitudes: as cosmic rays collide with atmospheric nuclei, cascades of secondary particles are produced. These showers peak in intensity at approximately **15–25 km** in a phenomenon known as the **Pfotzer maximum**. In this regime, the air density requirement to drive cascade effects **balances** with a minimum of overlying atmosphere.

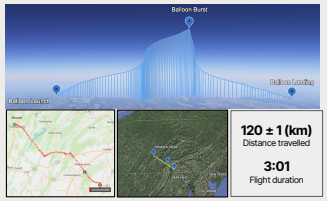
**JHNSRE is an inexpensive, high-altitude balloon apparatus built to characterize this maximum, and to photograph Earth from the middle stratospheric region:**




1. **MightyOhm Geiger PCB** — SBM-20 tube, 1.5V AAA, 400V HV+ lead
2. **Raspberry Pi Zero 2W** — OpenBSD 7.8, 32GB SD, gpspipe logs
3. **Camera + Ribbon Cable** — Raspberry Pi Camera Module 3W
4. **Molicel P28A 18650 ×2** — 5V series, Li-Ion, UPS HAT, to Raspberry Pi
5. **Enclosure, Satellite Antenna** — 2x GPS trackers, total mass 699.9 ±0.1 (g)
6. **Kaymont 1600 Balloon** — 4400L He, 2500g chute, ascent rate 5.6 m/s

JHNSRE was launched from Altoona, PA on 2026/04/24 following FAA clearance (#ZOB04-622) with the help of ~100 K-5 Bishop Guilfoyle Academy students. **The launch vehicle performed as expected**, traveling through a novel corridor over the Appalachian mountains. The trajectory taken closely followed pre-flight models developed by Cambridge University, with payload recovery near Gettysburg, PA.

**Result: Peak altitude attained = 35,500 ± 150 (m)**
Expected value = 35,000 ± 500 (m) (exact) // t = 0.96


Balloon Burst
Balloon Landing
Launch Site
Burst Point
Landing Point
Flight Path
Ground Track
Altitude (m)
Time (s)
Speed (m/s)
Vertical Speed (m/s)
Latitude
Longitude